

GLENCOE, MN, USA

WMO: 727517

Lat: 44.756N

Lon: 94.081W

Elev: 992

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04932

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-7.6	-19.3	2.0	-10.2	-14.9	2.5	-6.4	26.7	18.5	24.5	17.1	5.6	300	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	89.7	74.0	86.4	71.7	83.9	70.4	77.2	85.9	75.1	82.9	73.1	80.6	9.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.5	134.0	82.6	72.5	124.8	80.5	70.5	116.4	78.4	41.3	85.4	39.2	83.0	37.4	80.7	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	19.7	17.5	-18.1	95.1	6.5	2.5	-22.7	96.9	-26.5	98.4	-30.2	99.8	-34.9	101.6
			-18.5	80.5	6.3	2.4	-23.1	82.2	-26.8	83.6	-30.3	84.9	-35.0	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	15.6	18.7	32.4	46.5	58.1	68.4	72.4	69.3	62.6	48.2	35.2	20.9	
	DBStd	22.25	13.11	12.38	12.80	10.10	9.00	6.21	5.39	5.54	8.42	9.42	10.75	11.89	
	HDD50	4220	1065	876	560	180	26	0	0	0	7	149	455	902	
	HDD65	7645	1530	1296	1012	556	254	34	7	22	145	529	893	1367	
	CDD50	2702	0	0	14	77	276	553	696	597	384	93	12	0	
	CDD65	652	0	0	1	3	39	137	238	154	72	8	0	0	
	CDH74	5716	0	0	4	58	444	1176	2114	1193	633	93	1	0	
	CDH80	1690	0	0	0	14	148	353	672	305	177	21	0	0	
Wind	WSAvg	7.3	8.0	8.1	8.0	8.9	8.1	6.6	5.3	4.9	6.2	7.6	8.0	7.7	
Precipitation	PrecAvg	30.0	0.6	0.6	1.5	2.7	3.7	4.8	3.8	4.1	3.1	2.3	1.8	0.7	
	PrecMax	39.2	1.5	1.1	3.9	7.0	6.9	9.1	8.1	7.6	7.0	5.9	5.1	2.1	
	PrecMin	20.8	0.0	0.0	0.2	0.5	1.1	2.5	1.3	0.4	0.4	0.4	0.1	0.2	
	PrecStd	4.9	0.4	0.4	0.9	1.4	1.5	1.7	1.8	1.8	1.8	1.5	1.5	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	52.9	69.6	81.0	90.7	91.4	92.8	90.5	89.9	82.3	68.7	49.8	
		MCWB	39.2	50.0	58.7	58.7	67.2	73.0	77.9	75.3	72.4	63.8	56.6	46.5	
	2%	DB	38.0	43.2	61.4	72.9	84.0	87.8	89.7	86.2	84.5	75.5	61.2	42.7	
		MCWB	35.2	38.5	53.2	54.0	64.2	71.0	76.5	74.1	69.7	59.8	52.6	38.8	
	5%	DB	35.7	38.3	55.5	68.3	79.0	84.1	86.3	83.1	81.4	69.4	55.1	38.0	
		MCWB	34.1	35.2	50.7	53.7	62.6	68.7	73.8	71.5	67.7	57.8	48.6	35.8	
	10%	DB	32.7	35.5	49.9	63.3	73.4	81.0	83.7	80.9	77.2	63.4	50.2	35.9	
		MCWB	31.6	33.3	45.5	51.3	59.9	66.6	71.9	69.4	66.0	54.5	44.9	34.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	49.8	61.6	62.2	71.9	77.9	81.1	78.4	76.7	68.8	59.1	47.9
			MCDB	44.1	53.4	66.6	75.4	84.5	87.8	89.2	86.8	84.2	75.0	65.1	49.5
2%		WB	35.8	39.2	55.5	58.9	68.3	74.2	78.1	75.6	73.1	62.9	54.0	39.0	
		MCDB	37.7	42.4	58.9	68.4	78.2	83.3	86.7	83.7	79.7	72.8	59.6	41.5	
5%		WB	33.7	35.6	50.2	55.6	65.4	71.2	75.8	73.6	70.3	58.9	49.3	36.2	
		MCDB	35.2	38.1	56.0	64.7	74.8	80.0	83.6	80.9	77.8	67.6	55.0	38.1	
10%		WB	31.8	32.9	45.1	52.5	62.5	68.9	74.0	71.5	67.6	55.1	45.2	34.0	
		MCDB	32.8	34.8	49.4	62.0	70.7	77.0	81.4	78.0	75.7	62.2	50.0	35.5	
Mean Daily Temperature Range		5% DB	MDBR	15.1	16.4	15.9	19.0	19.9	18.7	19.2	19.2	20.9	18.9	16.0	13.7
			MCDBR	16.3	17.8	23.7	28.1	27.6	23.3	21.0	21.5	24.7	28.1	23.9	16.2
	MCWBR		14.0	14.3	16.2	15.5	12.8	10.4	10.7	11.1	12.1	15.3	15.8	13.8	
	5% WB	MCDBR	15.3	16.9	22.5	24.2	22.3	19.9	19.2	19.3	21.1	23.2	23.0	14.7	
		MCWBR	13.3	13.8	16.1	15.1	11.8	11.2	11.0	11.3	12.3	14.2	16.0	12.7	
Clear-Sky Solar Irradiance	taub		0.261	0.282	0.310	0.362	0.393	0.396	0.403	0.405	0.373	0.319	0.286	0.263	
	taud		2.449	2.380	2.425	2.336	2.308	2.345	2.333	2.312	2.370	2.521	2.521	2.458	
	Ebn at Noon		278	289	294	285	278	277	273	268	268	272	266	265	
	Edh at Noon		22	29	31	38	40	39	39	38	33	25	21	20	
All-Sky Solar Radiation	RadAvg		561	899	1187	1463	1700	1895	2030	1686	1320	850	546	403	
	RadStd		66	99	116	146	136	150	75	128	96	114	52	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page